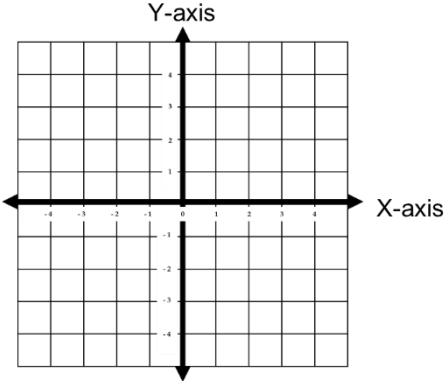
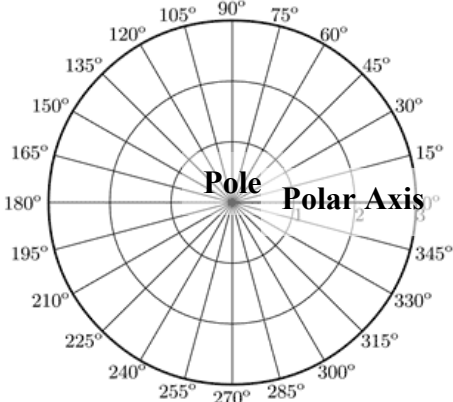


5.2 Polar Coordinates

<i>Cartesian (Rectangular) Coordinates</i>	<i>Polar Coordinates</i>
- Plot points by moving - horizontally (x-direction) and - vertically (y-direction)  - We have coordinates (x, y)	- Plot points by - rotating by a given angle and - moving a specified length  - We have coordinates (r, θ)

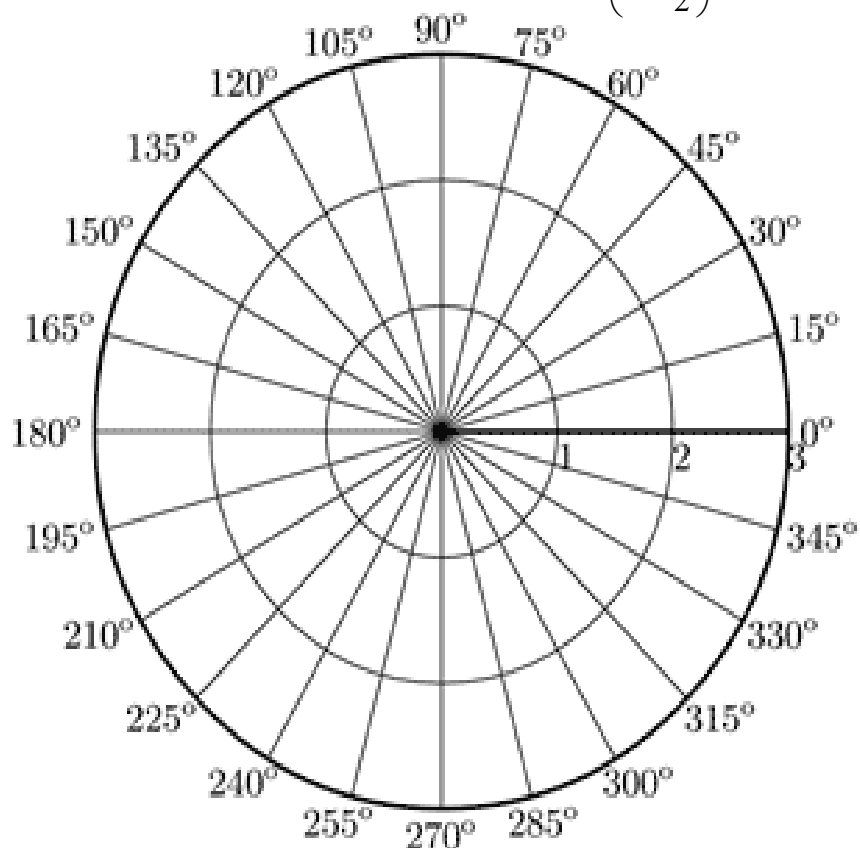
Example 1: Plot the points in the polar coordinate system provided below.

a. $A \left(2, \frac{5\pi}{6} \right)$

b. $B (-3, \pi)$

c. $C \left(1, -\frac{\pi}{2} \right)$

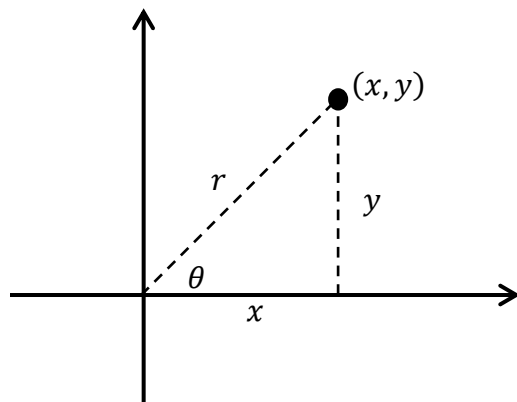
d. $D (-1, 450^\circ)$



Note: There are infinitely many ways to write the coordinates of a point in polar form!!!

To form the same point (r, θ) with different coordinates we simply use an angle that is coterminal with θ .

Polar –Rectangular Conversions



Converting from polar to rectangular

Converting from rectangular to polar

Example 2: Convert $(6, 210^\circ)$ to rectangular coordinates.

Example 3: Convert $(-3, 6)$ to polar coordinates.

Converting Equations

Formulas for converting equations:

$$r^2 = x^2 + y^2 \qquad x = r \cos(\theta) \qquad y = r \sin(\theta)$$

Common Techniques:

- (1) Multiplying both sides of the equation by r .
- (2) Squaring both sides of the equation.

Example 4: Write an equivalent rectangular equation for the polar equation: $r = 2 \cos(\theta)$.

Example 5: Use your calculator to sketch a graph of the polar curve given by $r = 1 + \sin(\theta)$.

Example 6: Use your calculator to sketch a graph of the polar curve given by $r = \cos(2\theta)$.

Example 7 (you try):

- (a) Convert the point $\left(2, \frac{\pi}{3}\right)$ from polar to rectangular (Cartesian) coordinates.
- (b) Represent the point with rectangular coordinates $(1, -1)$ in terms of polar coordinates.
- (c) Use your calculator to graph $r = \cos(\theta)$ and sketch the graph your calculator produces.
- (d) Convert the equation $r = \cos(\theta)$ from polar to rectangular form.